

Recall the proof that the halting problem is undecidable:

Solution: For the sake of contradiction, suppose there is a Turing machine H that decides the language $\text{HALT} = \{\langle M, w \rangle : M \text{ halts on input } w\}$. Now, we can decide the language $\text{SELFHALT} = \{\langle M \rangle : M \text{ halts on input } \langle M \rangle\}$ by running H on input $\langle M, \langle M \rangle \rangle$. We now show that this method decides SELFHALT correctly by showing that it correctly accepts and rejects the appropriate inputs.

- If M halts on $\langle M \rangle$, then H accepts on input $\langle M, \langle M \rangle \rangle$, and so our method accepts on input $\langle M \rangle$.
- If M does not halt on input $\langle M \rangle$, then H rejects on input $\langle M, \langle M \rangle \rangle$, and so our method rejects on input $\langle M \rangle$.

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- Below is the start of a proof that the language $\text{ACCEPTGRIZ} := \{\langle M \rangle \mid M \text{ accepts the string GRIZ}\}$ is undecidable.

Solution: For the sake of argument, suppose there is an algorithm DECIDEACCEPTGRIZ that correctly decides the language ACCEPTGRIZ . Then we can solve the halting problem as follows:

<u>$\text{DECIDEHALT}(\langle M, w \rangle)$:</u> Encode the following Turing machine M': <u>$M'(x)$:</u> <i>⟨ignore the input string x⟩</i> run M on input w <i>⟨ignore the output of M⟩</i> return TRUE if $\text{DECIDEACCEPTGRIZ}(\langle M' \rangle)$ return TRUE else return FALSE

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- List the four distinct Turing machines that are at play in this reduction and describe their role.
 - Prove that this reduction is correct by proving both
 - if M halts on input w , then DECIDEHALT accepts on $(\langle M, w \rangle)$, and
 - if DECIDEHALT accepts on $(\langle M, w \rangle)$, then M halts on input w . (Why is this a valid way to prove the correctness of the reduction?)
- Prove that each of the following languages is undecidable. For each proof, list *all* the Turing machines that are at play in the reduction and describe their role. To complete the proof, you must also prove the correctness of the reduction by proving that the TM you constructed to decide an undecidable language correctly decides that language.

- $\text{ACCEPT} = \{\langle M, w \rangle : M \text{ accepts } w\}$

(b) $\text{ALWAYSREJECT} = \{\langle M \rangle : \text{REJECT}(M) = \Sigma^*\}$